

Final Year Project Proposal

Outlook Integrated Smart AI Project Management Dashboard

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1. Introduction

Project managers handle many emails every day. These emails often include meeting schedules, report deadlines, pending tasks, required inputs, and follow-up actions. Manually tracking all this information takes time and increases the risk of missing important deadlines.

This project proposes a smart AI-based project management dashboard integrated with Microsoft Outlook. The system will read emails and their attachments, extract important project-related information, store it in Microsoft Excel, and show it on a centralized dashboard. It will help project managers track meetings, deadlines, action items, and follow-ups in an organized way, while respecting the different access needs of managers at different levels of the organisation.

2. Problem Statement

In many organizations, project-related information is shared through emails. Important details like meeting dates, report deadlines, required documents, and pending inputs are often hidden inside long email threads and in the files attached to them.

Project managers usually track these details manually using Excel sheets, notes, or reminders. This process is time-consuming and error-prone. Missed follow-ups can delay project tasks and affect team performance. Adding to the difficulty, different people in the same organisation, such as a senior manager, a project manager, and a team member, need different views of the same information, but most manual tracking methods make no such distinction.

There is a need for a smart system that can automatically detect useful information from Outlook emails and their attachments, store it properly, present it according to each user's role in the organisation, and generate reminders before deadlines.

3. Proposed Solution

The proposed system will work as an AI email assistant for project managers. It will connect with Outlook 2016 and identify emails related to meetings, deadlines, action items, and required inputs.

The system will extract key details such as:

- Email subject
- Meeting date and time
- Participants
- Meeting agenda
- Deadlines
- Required documents
- Action items

- Responsible department or person

In addition to the email body, the system will extract information directly from attachments, such as Word documents, PDFs, and scanned images, so that details inside a Minutes of Meeting file, a report, or a presentation are captured even when the email text itself is brief.

The system will also recognise that Outlook is used differently depending on where a person sits in the organisation. A senior manager may need a consolidated view across several projects and departments, a project manager may need a focused view of their own project's meetings and deadlines, and a team member may only need to see the action items assigned to them. The system will apply different extraction and display rules depending on the user's position in this hierarchy, rather than presenting the same undivided view to everyone.

After extraction, the system will store the information in Microsoft Excel for record keeping. The data will also appear on a centralized dashboard where the project manager can view meetings, tasks, alerts, pending inputs, and completed actions.

The system will also send automatic reminders before deadlines and ask for post-meeting documents such as minutes of meeting, reports, presentations, and pending updates.

4. Aim of the Project

The main aim of this project is to develop a smart project management dashboard that helps project managers track emails, meetings, deadlines, and follow-ups automatically through Outlook and Excel integration, with extraction that covers both email text and attachments, and with views suited to each user's place in the organisational hierarchy.

5. Objectives

The main objectives of this project are:

- To develop a stand-alone software system for project tracking.
- To integrate the system with Microsoft Outlook 2016.
- To detect project-related emails automatically.
- To extract meeting details, deadlines, participants, and action items from emails.
- To extract relevant information from email attachments, including Word documents, PDFs, and scanned images.
- To store extracted data in Microsoft Excel.
- To display all records in a centralized dashboard.
- To generate reminders and alerts before deadlines.
- To track pending inputs, reports, presentations, and meeting documents.

- To support differentiated Outlook-based access and dashboard views according to organisational hierarchy, such as senior manager, project manager, and team member.
- To support single-user and multi-user usage within a local network.
- To ensure the system operates entirely in the English language.

6. Scope of the Project

The scope of this project includes the development of a desktop-based smart project management system. The system will work with Outlook 2016 for email reading and Microsoft Excel for data storage and archival.

The system will focus on project-related emails and will help project managers manage their workflow without changing their current working style. It will support basic AI and NLP techniques along with rule-based logic to understand semi-structured and unstructured email content, and will extend this understanding to common attachment formats rather than email text alone.

The system will also account for the fact that Outlook is used differently across different levels of an organisation. Access rules and dashboard views will differ depending on whether the user is a senior manager, a project manager, or a team member, so that each user sees information relevant to their role.

The project will be developed and operated entirely in the English language. Email content, extracted data, and the dashboard interface will be designed for English-language input and output only; support for other languages is not part of this project's scope.

7. Key Features

7.1 Outlook Email Detection

The system will scan Outlook emails and identify emails related to meetings, deadlines, tasks, and follow-ups.

7.2 Information Extraction

The system will extract important details from emails, including subject, date, time, participants, agenda, action items, and required documents.

7.3 Attachment-Based Information Extraction

The system will open and read common attachment types, including Word documents, PDFs, and scanned images, and extract relevant project details from them, such as meeting minutes, deadlines, and action items, and link this information to the email it was sent with.

7.4 Excel-Based Data Logging

All extracted information, from both email bodies and attachments, will be stored in Microsoft Excel for record keeping, reporting, and future reference.

7.5 Centralized Dashboard

The dashboard will show meetings, upcoming deadlines, pending tasks, completed tasks, and required follow-ups in one place, with the specific view adjusted to the signed-in user's role.

7.6 Reminder and Alert System

The system will generate alerts before important deadlines and remind users about pending inputs and post-meeting documents.

7.7 Custom Settings

Users will be able to manage reminder timing, tracking preferences, alert settings, and other basic options.

7.8 Hierarchy-Aware Outlook Access

The system will support different Outlook access and dashboard configurations depending on a user's position in the organisational hierarchy. Senior managers will be able to see a consolidated view across projects and departments, project managers will see their own project's details, and team members will see only the items assigned to them.

7.9 Local Network Support

The software will support single-user and multi-user use within a local network.

8. Methodology

The system will follow a hybrid approach. It will use lightweight AI and Natural Language Processing techniques along with rule-based logic.

The methodology includes the following steps:

1. Email and Attachment Collection

The system will connect with Outlook 2016 and collect email data along with any attached files.

2. Email Classification

The system will identify whether an email is related to a meeting, deadline, report, task, or follow-up.

3. Information Extraction

NLP and rule-based methods will extract important details from the email text and from supported attachment formats.

4. Hierarchy-Based Routing

Extracted records will be tagged according to the sender's and recipient's position in the organisational hierarchy, so that later views can be filtered by role.

5. Data Storage

The extracted data will be saved in Microsoft Excel for archival and tracking.

6. Dashboard Visualization

The stored data will be displayed in a clear dashboard suited to each user's role.

7. Alert Generation

The system will generate reminders before deadlines and follow-up dates.

8. User Customization

Users will be able to change settings based on their workflow.

9. Tools and Technologies

The expected tools and technologies for this project are:

- Programming Language: Python
- Email Platform: Microsoft Outlook 2016
- Data Storage: Microsoft Excel
- AI/NLP Libraries: spaCy, NLTK, or Scikit-learn
- Attachment Parsing: PyPDF2 / pdfplumber (PDF), python-docx (Word), Pillow and pytesseract (OCR for scanned images)
- Dashboard: Tkinter, PyQt, Streamlit, or Flask
- Automation: Python libraries for Outlook and Excel integration
- Local Network Support: Basic client-server or shared file-based access

10. Expected Deliverables

The expected deliverables of this project are:

- A stand-alone software system with a centralized dashboard.
- Outlook 2016 email detection and reading module.
- Attachment reading and extraction module for Word, PDF, and image files.
- Microsoft Excel data logging and archival system.
- AI/NLP-based email and attachment information extraction module.
- Rule-based email classification and deadline detection.
- Hierarchy-aware access and dashboard configuration.
- Reminder and alert generation system.

- Custom settings for input tracking and alerts.
- Dashboard for meetings, deadlines, tasks, and pending inputs.
- Final documentation and project report.

11. Key Performance Indicators

The success of the project will be measured through the following KPIs:

- Accuracy of email type detection.
- Accuracy of extracted meeting and deadline details.
- Accuracy of information extracted from attachments.
- Correctness of role-based access and dashboard views across the organisational hierarchy.
- Successful data logging in Microsoft Excel.
- Timely generation of alerts and reminders.
- Dashboard clarity and ease of use.
- Reduction in manual tracking effort.
- Support for single and multiple users within a local network.

12. Job Descriptions

Project responsibilities are divided among the three team members as follows:

Zain Ahmed Siddiqui: Outlook Integration & Backend Lead

- Build and maintain the Outlook 2016 connectivity module, including reading emails, threads, and attachments.
- Design the hierarchy-aware access rules that determine what each user role can see.
- Develop the Microsoft Excel data logging and archival module.
- Coordinate integration between the email/attachment pipeline and the data storage layer.

Syed Faaiz Noman: AI/NLP Lead

- Design and implement the email and attachment classification models.
- Build the NLP and rule-based information extraction pipeline for email bodies and attachments (PDF, Word, Excel, etc).
- Develop the reminder and alert generation logic, including deadline detection and follow-up tracking.
- Evaluate extraction and classification accuracy and tune the models accordingly.

Zoha Asim: Dashboard, Testing & Documentation Lead

- Design and build the centralized dashboard, including role-specific views for different hierarchy levels.
- Implement the custom settings module for reminders, tracking preferences, and alerts.

- Lead testing and quality assurance across the extraction, storage, and dashboard modules.
- Prepare and maintain project documentation, including the final report and presentation materials.

13. Group Projects

NLP Studio: Intelligent Text Analysis Web Application: A browser-based NLP suite with grammar checking, summarization, NER, sentiment analysis, and BERT-based autocomplete/paraphrasing.

Text Emotion Classifier using Bi-LSTM: An NLP pipeline classifying text emotions, achieving 92% accuracy with Bi-LSTM and SHAP interpretability.

Library Management System: A system for managing book records, memberships, and borrowing/return transactions.

Human Resource Management System: A system for managing employee records, attendance, and HR processes.

Event Management System: A system for planning, scheduling, and tracking events and registrations.

14. Expected Outcome

The final system will help project managers manage emails, meetings, deadlines, and follow-ups in a better way. It will reduce manual work, improve task visibility, and help teams complete work on time.

The system will also maintain a formal record of meetings, deadlines, reports, and pending inputs, drawn from both email bodies and attachments. This will help senior managers track project progress at the level of detail appropriate to their role, without changing their existing management style.

15. Conclusion

The Outlook Integrated Smart AI Project Management Dashboard is a useful project for improving project tracking and email-based task management. It combines Outlook, Excel, AI/NLP, and a centralized dashboard to help project managers work more efficiently.

By extracting information from both emails and their attachments, and by adapting to different levels of the organisational hierarchy, the system will save time, reduce missed deadlines, and create a structured, English-language record of all important project activities.